Course Name: Electrical and Electronics Engineering Lab
Course Code: EE-102

Contact Hours/Week: **2P** Course Credits:**01**

Course Objectives

- To impart basic knowledge of electrical quantities such as current, voltage, power, energy etc.
- To familiarize students with basic circuit components and their connections.
- To explain working principle of electrical measuring instruments such as ammeter, voltmeter, wattmeter, energy meter etc.
- Familiarization with electronic components and equipments.
- To validate and verify the characteristics of various electronic devices.
- To implement electronic circuits using different electronic components.

List of Experiments

1. To verify fundamental laws of Electrical Engineering
    I. Ohm's law for Bilateral Passive Linear Lumped (BPLL) element.
    II. KVL and KCL
2. To verify Network theorems
    I. Norton and Thevenin Theorem
    II. Maximum power theorem
    III. Superposition theorem
    IV. Tellegen's Theorem
3. To calibrate a measuring instrument
    I. Voltmeter with the help of standard ammeter and resistance.
    II. Ammeter with the help of standard voltmeter and resistance.
    III. Wattmeter by direct loading.
4. To find minimum fusing current and fuse constant of a given fuse wire.
5. To check polarity markings of a transformer and to determine self & mutual inductance of windings.
6. To find voltage current relationship in R-L series circuit and to determine power factor of the circuit. Also determine the inductance of a coil using voltmeter, ammeter methods
7. Familiarization of electronic components and equipments like CRO, function generator and power supplies etc.
8. To study the V-I characteristics of p-n junction diode and determine its static and dynamic resistance.
9. To study the characteristics of Zener diode, calculate the dynamic resistance and working as Voltage regulator.
10. To study and plot the waveform of half wave and full wave rectifier with and without capacitor filter.
11. To study and plot the input and output characteristics of CE (Common Emitter) and CB (Common Base) transistor configurations and calculate their input and output resistances.
12. To study the characteristics of FET (Field Effect Transistor) and calculate its dynamic resistance (rd), mutual conductance (gm) and amplification factor (μ).

***NOTE:** The experiments of EE-102 course shall be jointly conducted by Electrical Engineering and Electronics & Communication Engineering Departments. However, the final grade submission of the Lab course EE-102 shall be done by Electrical Engineering Department.

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Verify fundamental laws like Ohm's law, KCL, KVL and use different meters and instruments for the measurement of common electrical quantities.

CO2: Understand the importance of fuse as a safety device and study the parameters related to the selection of fuse wire.

CO3: The basic working principle & construction of a transformer.

CO4: Understanding of different meters and instruments for measurement of electronic quantities.

CO5: Develop skills for designing electronics circuits and its practical implementation on breadboard.

CO6: Understand the characteristics of different electronic devices like diodes, BJT and FET.